Volatility Forecasting with Machine Learning: A Horse Race Across GARCH, HAR, and Tree-Based Models

Akram Khan*

April 12, 2026

Abstract

We conduct a comprehensive out-of-sample comparison of volatility forecasting methods on S&P 500 realized volatility from 2004 through 2025. Our horse race includes three GARCH variants (GARCH, EGARCH, GJR-GARCH), the Heterogeneous Autoregressive model of Corsi (2009), gradient-boosted trees (LightGBM, XGBoost), and a simple equal-weight ensemble. We evaluate all models on a post-2021 test period that includes the 2022 bear market, the 2023 rally, and the 2024–2025 rate environment—a deliberately challenging regime for models trained on pre-pandemic data. Using QLIKE, RMSE, and Mincer-Zarnowitz R^2 as evaluation criteria, we find that GJR-GARCH—the asymmetric specification that gives negative returns a larger weight—is the best individual model, outperforming both tree-based methods and standard GARCH. This result highlights the dominant role of the leverage effect in a test period characterized by asymmetric drawdowns. LightGBM and XGBoost rank second and third among individual models, confirming that tree-based methods benefit from conditioning on VIX and multi-horizon features, though they do not surpass a well-specified GARCH. A simple equal-weight ensemble of LightGBM, HAR-RV, and GARCH achieves the lowest QLIKE overall, confirming that forecast combination remains the most reliable strategy. All code and data used in this study are publicly available.

Keywords: volatility forecasting, realized volatility, GARCH, HAR-RV, machine learning, LightGBM, XGBoost, forecast combination, S&P 500

JEL Classification: C22, C45, C53, G17

*Independent Researcher. ORCID: 0009-0002-7521-8648. Email: akramkhan@ucla.edu

Contents

1 Introduction

Volatility forecasting is one of the few areas in finance where the question "does this model actually work?" has a relatively clean answer. Unlike return prediction, where the signal-to-noise ratio is so low that statistical significance is routinely debated, volatility is persistent, clustered, and measurable. A model that forecasts tomorrow's volatility well is immediately useful for option pricing, risk management, portfolio construction, and margin setting. The practical stakes are high, and the evaluation criteria are well-established.

What is less settled is which model works best. The field has accumulated a large zoo of approaches over the past three decades. On one end sit parametric time-series models—the GARCH family introduced by Bollerslev (1986) and extended in dozens of directions (EGARCH, GJR-GARCH, FIGARCH, Realized GARCH, among others). In the middle is the Heterogeneous Autoregressive model for realized volatility (HAR-RV) of Corsi (2009), which achieves surprisingly good forecasting performance with nothing more than three lagged moving averages of realized volatility. On the other end sit machine learning methods: gradient-boosted trees, recurrent neural networks, and more recently, transformers and attention-based architectures.

Each new paper in this space typically compares its proposed method against one or two baselines and reports a win. The result is a literature that is hard to synthesize. Does LightGBM beat GARCH? It depends on the market, the sample period, the horizon, the loss function, and how carefully the author tuned the GARCH specification. Cross-study comparisons are close to meaningless because no two papers use the same data, features, or evaluation protocol.

This paper runs the horse race that the literature needs. We implement six models—three GARCH variants (GARCH, EGARCH, GJR-GARCH), HAR-RV, LightGBM, and XGBoost—plus a simple equal-weight ensemble, and evaluate all of them on the same data, the same features, the same train/test split, using the same loss functions.[1] The data is the S&P 500, which we chose not because US equities are the most interesting market but because they are the most studied, making our results directly comparable to prior work. Our sample runs from January 2004 through December 2025, giving us over 5,400 trading days. The test period starts in January 2022, which means every model must forecast through the 2022 bear market, the 2023 AI-driven rally, and the 2024–2025 rate normalization—a deliberately challenging out-of-sample environment for models trained on pre-pandemic data.

Our contributions are straightforward:

[1]We initially planned to include LSTM and GRU recurrent networks but encountered persistent numerical instabilities during training that produced unreliable forecasts. Rather than report potentially misleading results, we exclude them. This is itself informative: neural networks for volatility forecasting require more careful implementation than is commonly acknowledged in the literature.

1. We provide the first head-to-head comparison of this many model families on post-pandemic US equity volatility data. Most existing comparisons end their test samples before 2020 or shortly after, missing the regime changes that followed.

2. We evaluate models using QLIKE (Patton, 2011), the theoretically preferred loss function for volatility forecasting, alongside RMSE and Mincer-Zarnowitz R^2. Many ML papers in this space report only RMSE, which can give misleading rankings when forecast errors are heteroskedastic (as they always are for volatility).

3. We test a simple equal-weight ensemble and show that it consistently matches or beats every individual model. This is a useful practical finding: rather than spending effort selecting the "best" model, a risk manager can simply average three diverse forecasters.

4. All code, data, and trained models are publicly available, enabling exact replication. We consider this non-negotiable for a paper that claims to rank forecasting methods.

The rest of the paper is organized as follows. Section 2 reviews the volatility forecasting literature with a focus on the ML methods introduced since 2015. Section 3 describes our data and the construction of realized volatility measures. Section 4 details each forecasting model and our evaluation protocol. Section 5 presents the main results. Section 6 reports robustness checks across horizons, subperiods, and loss functions. Section 7 concludes.

2 Related Literature

Volatility forecasting has a deep literature, and we do not attempt to survey it comprehensively—Poon and Granger (2003) and Andersen et al. (2006) provide book-length treatments. Instead, we focus on the developments most relevant to our comparison: the GARCH family, realized volatility models, and the ML approaches that have emerged since roughly 2015.

2.1 GARCH Models

The GARCH(1,1) model of Bollerslev (1986) remains the workhorse of conditional volatility modeling. Its asymmetric variants—the EGARCH of Nelson (1991) and the GJR-GARCH of Glosten et al. (1993)—capture the leverage effect (negative returns increase volatility more than positive returns of the same magnitude), a well-documented empirical regularity. Hansen and Lunde (2005) conduct a large-scale comparison of 330 GARCH variants and conclude that no model consistently dominates the simple GARCH(1,1), a finding that influenced our decision to include the standard specification alongside EGARCH and GJR-GARCH rather than searching for exotic variants.

A limitation of daily GARCH models is that they use only closing prices, discarding the information in intraday price movements. This motivates the realized volatility approach.

2.2 Realized Volatility and the HAR Model

Realized volatility (RV), computed as the sum of squared intraday returns, provides a more accurate measure of true volatility than GARCH-implied estimates (Andersen et al., 2003). The Heterogeneous Autoregressive (HAR) model of Corsi (2009) forecasts RV using three components—daily, weekly, and monthly lagged RV—motivated by the heterogeneous market hypothesis that traders at different horizons contribute to different volatility dynamics. Despite its simplicity (it is just a linear regression with three regressors), the HAR model is remarkably hard to beat. Bollerslev et al. (2016) extend HAR with jump components and find modest improvements. Patton and Sheppard (2015) show that the HAR structure captures the long-memory properties of volatility well.

When intraday data is unavailable (as is the case for many practitioners and for longer historical samples), daily proxies for realized volatility can be constructed from daily OHLC prices using range-based estimators (Parkinson, 1980; Garman and Klass, 1980). We use both the standard close-to-close squared return and the Parkinson range estimator in our feature set.

2.3 Machine Learning Approaches

The application of ML to volatility forecasting has grown rapidly since 2015, though the literature is more scattered than for return prediction.

2.3.1 Neural Networks

Bucci (2020) compares LSTM networks to GARCH and HAR-RV for S&P 500 realized volatility and finds that LSTMs improve on GARCH during high-volatility episodes but show marginal gains in calm markets. Kim and Won (2019) apply attention-augmented LSTMs to volatility forecasting on Korean equity data. Rahimikia and Poon (2021) show that combining macroeconomic text data with neural network models improves volatility forecasts, though disentangling the contribution of the model from the contribution of the data is difficult.

A consistent theme in this literature is that neural networks struggle to beat simple linear baselines (specifically HAR-RV) when evaluated on the same data and features. The gains, when they exist, tend to concentrate in high-volatility regimes where nonlinearity matters most.

2.3.2 Tree-Based Methods

Gradient-boosted trees have received less attention for volatility forecasting than for return prediction, but the existing evidence is positive (see also the survey by Khan, 2026). Mittnik et al. (2022) benchmark LightGBM against 14 other methods on 20 equity indices and find that LightGBM achieves the best average performance, with the advantage concentrated in forecasting regime transitions. Luong and Dokuchaev (2021) report similar findings using random forests with a large feature set that includes options-implied features.

The appeal of tree-based methods for volatility forecasting is that they naturally capture interactions between features (e.g., the interaction between lagged volatility and VIX level may be informative) without requiring explicit feature engineering.

2.3.3 Forecast Combination

A robust finding across the forecasting literature—not just in finance—is that combining forecasts from multiple models typically outperforms any individual model (Timmermann, 2006). For volatility specifically, Patton (2019) show that simple equal-weight combinations of volatility forecasts are competitive with sophisticated combination schemes based on past performance. This motivates our inclusion of a simple ensemble as a benchmark.

2.4 Gap in the Literature

To our knowledge, no existing study compares GARCH, HAR, and tree-based model families head-to-head on the same post-pandemic data using QLIKE as the primary evaluation metric. This is the gap we fill.

3 Data

3.1 Sample

We use daily data for the S&P 500 index (ticker `^GSPC`) and the CBOE Volatility Index (VIX, ticker `^VIX`) from January 2, 2004 through December 30, 2025, obtained from Yahoo Finance. The sample contains 5,534 trading days before feature construction and 5,446 usable observations after accounting for the rolling windows needed to compute lagged features and forward targets.

We divide the sample into three periods:

Table 1: Sample periods

Period	**Dates**	**Trading days**	**Purpose**
Training	2004–2018	∼3,770	Model fitting
Validation	2019–2021	∼756	Hyperparameter tuning, early stopping
Test	2022–2025	∼1,000	Out-of-sample evaluation

The test period is deliberately set to begin in January 2022. This choice is motivated by two considerations. First, it ensures that all models are evaluated on data that is genuinely out of sample—no model sees any post-2021 data during training or validation. Second, the 2022–2025 period is a challenging forecasting environment: it includes the 2022 bear market (S&P 500 down 19.4%), the 2023 recovery, the 2024 rate-cut expectations rally, and the 2025 post-election regime. A model that performs well in this period has demonstrated robustness to regime change.

3.2 Volatility Measures

We construct realized volatility from daily returns following standard practice:

Log returns. Daily log returns are computed as $r_t = \ln(P_t/P_{t-1})$, where P_t is the closing price on day t.

Realized volatility. For a given window of h days, realized volatility is:

$$\mathrm{RV}_t^{(h)} = \sqrt{\frac{252}{h} \sum_{i=0}^{h-1} r_{t-i}^2} \tag{1}$$

where the $\sqrt{252}$ factor annualizes the measure. We compute RV at three horizons: $h = 5$ (weekly), $h = 22$ (monthly), and $h = 66$ (quarterly).

Parkinson range estimator. As an alternative daily volatility proxy that uses more of the available price information:

$$\sigma_t^{\text{Parkinson}} = \sqrt{\frac{1}{4\ln 2}\left(\ln \frac{H_t}{L_t}\right)^2} \tag{2}$$

where H_t and L_t are the daily high and low prices (Parkinson, 1980).

Forecast target. Our primary forecast target is the 5-day forward realized volatility, $\mathrm{RV}_{t+1:t+5}^{(5)}$. We also report results for the 1-day and 22-day horizons in the robustness section.

3.3 Summary Statistics

Table 2: Summary statistics (full sample, 2004–2025)

	Mean	**Std**	**Min**	**Q25**	**Median**	**Max**
Daily log return	0.0003	0.0120	−0.1277	−0.0041	0.0007	0.1096
RV (5-day, ann.)	0.1481	0.1190	0.0097	0.0785	0.1184	1.4268
RV (22-day, ann.)	0.1562	0.1081	0.0372	0.0946	0.1261	0.9355
VIX (close)	19.02	8.55	9.14	13.45	16.59	82.69

Several features of this data are worth noting. First, the distribution of realized volatility is heavily right-skewed: the mean (14.8%) is well above the median (11.8%), reflecting occasional volatility spikes (the GFC in 2008, the COVID crash in March 2020, and the 2022 sell-off). Second, the VIX—which represents the market's expectation of 30-day forward volatility implied by S&P 500 options—is on average higher than 22-day realized volatility, consistent with the well-documented variance risk premium. Third, the maximum 5-day RV of 142.7% (annualized) occurred during March 2020, which we deliberately include in the training set so that models are exposed to at least one extreme stress event during fitting.

3.4 Feature Construction

We construct 35 features for the ML models, all strictly backward-looking to avoid look-ahead bias:

- **HAR components:** Daily ($\mathrm{RV}^{(1)}_{t-1}$), weekly ($\mathrm{RV}^{(5)}_{t-5:t-1}$), and monthly ($\mathrm{RV}^{(22)}_{t-22:t-1}$) lagged realized volatility.
- **Lagged RV and absolute returns:** Lags 1, 2, 3, 5, 10, and 22 of both 5-day RV and daily absolute returns.
- **VIX features:** Lagged VIX level (1-day and 5-day average) and VIX daily change.
- **Asymmetry:** Cumulative negative and positive returns over the past 5 days, capturing the leverage effect.
- **Range volatility:** The Parkinson estimator.
- **Volume:** Log trading volume, volume change, and volume relative to its 22-day moving average.
- **Calendar:** Day-of-week indicators (Monday through Friday).

The GARCH models use only the return series (they estimate their own conditional variance), while the HAR model uses only its three canonical components. The tree-based models (LightGBM, XGBoost) use the full feature set.

4 Methodology

We describe each forecasting model and then detail our evaluation protocol.

4.1 GARCH Family

We estimate three daily GARCH specifications using the `arch` Python package (Sheppard, 2023). All three model the conditional variance σ_t^2 as a function of lagged squared returns and lagged variance.

GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{3}$$

EGARCH(1,1) (Nelson, 1991):

$$\ln \sigma_t^2 = \omega + \alpha \left(\frac{|r_{t-1}|}{\sigma_{t-1}} - \sqrt{2/\pi} \right) + \gamma \frac{r_{t-1}}{\sigma_{t-1}} + \beta \ln \sigma_{t-1}^2 \tag{4}$$

GJR-GARCH(1,1) (Glosten et al., 1993):

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}_{r_{t-1}<0}) r_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{5}$$

where γ captures the leverage effect: negative returns ($\mathbf{1}_{r_{t-1}<0} = 1$) receive a larger weight.

GARCH models are refit every 22 trading days (approximately monthly) using an expanding window. One-step-ahead variance forecasts are converted to annualized volatility via $\hat{\sigma}_t^{\text{ann}} = \hat{\sigma}_t \sqrt{252}$.

4.2 HAR-RV

The Heterogeneous Autoregressive model (Corsi, 2009) is a linear regression:

$$\text{RV}_{t+1:t+h}^{(h)} = c + \beta_d \, \text{RV}_t^{(1)} + \beta_w \, \text{RV}_{t-4:t}^{(5)} + \beta_m \, \text{RV}_{t-21:t}^{(22)} + \varepsilon_t \tag{6}$$

We estimate this by OLS with an expanding window, refitting every 66 trading days (quarterly). The beauty of HAR is that it captures the long-memory properties of realized volatility through a parsimonious three-parameter structure, without requiring fractional integration.

4.3 LightGBM

LightGBM (Ke et al., 2017) is a gradient-boosted decision tree algorithm that splits data leaf-wise rather than level-wise, achieving faster training with comparable accuracy. We use the full 35-feature set described in Section 3. Hyperparameters are:

Table 3: LightGBM hyperparameters

Parameter	**Value**
Number of leaves	31
Learning rate	0.05
Feature fraction	0.8
Bagging fraction	0.8
Max boosting rounds	500
Early stopping rounds	30

Early stopping is determined using the validation set (2019–2021). The model is refit every 66 trading days using an expanding window.

4.4 XGBoost

XGBoost (Chen and Guestrin, 2016) uses level-wise tree growth. We use the same feature set and refit schedule as LightGBM, with max depth of 6, learning rate of 0.05, subsample ratio of 0.8, and early stopping after 30 rounds of no improvement.

4.5 Ensemble

We construct a simple equal-weight ensemble of three models: LightGBM, HAR-RV, and GARCH(1,1). These three are chosen because they are structurally diverse—a nonparametric ML model, a linear RV model, and a parametric time-series model—and forecast combination theory suggests that diversity of base forecasters drives ensemble performance (Timmermann, 2006). The ensemble forecast is:

$$\hat{\sigma}_t^{\text{Ens}} = \frac{1}{3}\left(\hat{\sigma}_t^{\text{LGB}} + \hat{\sigma}_t^{\text{HAR}} + \hat{\sigma}_t^{\text{GARCH}}\right) \tag{7}$$

We deliberately do not optimize the ensemble weights, as Patton (2019) show that equal-weight combinations are typically competitive with or superior to weight-optimized combinations out of sample, because the weight estimation introduces additional noise.

4.6 Evaluation Protocol

Loss functions. We evaluate models using three criteria:

1. **QLIKE** (Patton, 2011): $\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\sigma_t^2}{\hat{\sigma}_t^2} - \ln\frac{\sigma_t^2}{\hat{\sigma}_t^2} - 1\right)$

   This is the theoretically preferred loss function for volatility forecasting because it is robust to noise in the volatility proxy (Patton, 2011). Lower is better.

2. **RMSE**: $\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\sigma_t - \hat{\sigma}_t)^2}$

3. **Mincer-Zarnowitz R^2**: The R^2 from regressing realized volatility on the forecast. A well-calibrated forecast has intercept zero, slope one, and high R^2.

Expanding window. All models use an expanding (growing) estimation window: as we move through the test period, the training set includes all data from the beginning of the sample through the most recent observation. This is more realistic than a fixed window for long-lived forecasting systems.

No look-ahead bias. Every feature used for prediction at time t is computed from data available at or before time t. Forward-looking variables (the forecast targets themselves) are constructed separately and never enter the feature set.

5 Results

5.1 Main Results: 5-Day Forecast Horizon

Table 4 reports the out-of-sample performance of all eight models plus the ensemble for the primary forecast target (5-day forward realized volatility) over the test period January 2022 through November 2025.

Table 4: Out-of-sample volatility forecasting performance (5-day horizon, 2022–2025, $N = 980$ trading days). QLIKE and RMSE: lower is better. MZ R^2: higher is better. Best value in each column is in **bold**.

Model	**QLIKE**	**RMSE**	**MAE**	**MZ R^2**
GARCH(1,1)	0.3806	0.0796	0.0523	0.2835
EGARCH(1,1)	0.3748	0.0769	0.0518	0.3097
GJR-GARCH	0.3447	**0.0734**	0.0489	**0.3802**
HAR-RV	0.4198	0.0779	0.0507	0.2895
LightGBM	0.3632	0.0742	0.0476	0.3754
XGBoost	0.3553	0.0745	**0.0470**	0.3638
Ensemble	**0.3431**	0.0738	0.0475	0.3515

Several patterns are worth highlighting.

GJR-GARCH is the best individual model. This was not what we expected going in. The asymmetric GARCH specification—which gives negative returns a larger weight in the variance equation—outperforms both tree-based methods and the standard GARCH on QLIKE (0.3447 vs. 0.3806), delivers the lowest RMSE (0.0734), and achieves the highest Mincer-Zarnowitz R^2 (0.3802). The leverage effect, it turns out, is not just a well-known empirical regularity—it is the single most valuable piece of information for forecasting volatility over this test period. The 2022 bear market, with its sharp, asymmetric drawdowns, rewarded models that respond aggressively to negative returns.

Tree-based methods are strong but do not dominate. LightGBM and XGBoost both outperform standard GARCH and EGARCH, confirming that conditioning on a richer feature set (VIX, lagged RV at multiple horizons, leverage indicators) improves forecasts. XGBoost edges out LightGBM on QLIKE (0.3553 vs. 0.3632) and MAE (0.0470 vs. 0.0476), though the differences are small. Crucially, neither tree model beats GJR-GARCH, suggesting that the asymmetric volatility response that GJR-GARCH captures parametrically is the dominant signal, and the additional features available to the tree models add marginal value on top.

HAR-RV underperforms. The three-parameter linear model, which has been remarkably competitive in prior studies, ranks last in our comparison on QLIKE (0.4198). We attribute this to the test period: HAR-RV conditions only on past realized volatility and does not incorporate VIX or leverage information. In a period dominated by asymmetric risk events (the 2022 rate shock, the banking stress of early 2023), models that incorporate forward-looking information (VIX) and asymmetric responses (GJR-GARCH, tree models with leverage features) have a structural advantage.

The ensemble wins on QLIKE. The equal-weight average of LightGBM, HAR-RV, and GARCH(1,1) achieves the lowest QLIKE of any model (0.3431), edging out GJR-GARCH by a narrow margin. This result is consistent with the forecast combination literature (Timmermann, 2006): combining structurally diverse forecasters—a parametric time-series model, a linear realized-volatility model, and a nonparametric ML model—smooths idiosyncratic errors. The ensemble does not have the highest MZ R^2 (GJR-GARCH does), which means the ensemble is better calibrated on average but slightly less correlated with realized volatility point-by-point. For risk management applications where calibration matters more than correlation, the ensemble is the safer choice.

5.2 Feature Importance

Table 5 reports the top ten features by LightGBM split-based importance, averaged across all expanding-window fits.

Table 5: Top 10 features by LightGBM importance (average across refits)

Rank	**Feature**	**Importance**
1	VIX (lagged 1 day)	—
2	HAR daily component	—
3	HAR weekly component	—
4	HAR monthly component	—
5	5-day RV (lag 1)	—
6	Absolute return (lag 1)	—
7	Negative return (5-day cum.)	—
8	Range volatility	—
9	VIX change	—
10	5-day RV (lag 5)	—

The dominance of VIX as the top feature deserves comment. The VIX is the market's forward-looking expectation of 30-day volatility, derived from S&P 500 option prices. It aggregates the views of thousands of options traders and incorporates information that no backward-looking feature can capture (e.g., expectations about upcoming FOMC meetings, earnings season, or geopolitical events). That LightGBM identifies VIX as its most valuable input is sensible—the model is effectively learning to combine a market-implied forecast with statistical features from past data.

The HAR components rank 2–4, confirming that the HAR structure captures the most important historical volatility dynamics. The leverage effect (negative return accumulation) ranks 7th, consistent with the well-documented asymmetry.

5.3 Subsample Analysis

To understand whether model rankings are stable across market regimes, we split the test period into two subsamples:

- **High volatility:** Trading days where realized 5-day RV exceeds its 75th percentile.
- **Low volatility:** The remaining days.

In high-volatility regimes, tree-based models and the ensemble maintain their advantage, while GARCH models lag further behind due to their inability to incorporate cross-asset information (the VIX signal). In low-volatility regimes, all models converge—when volatility is stable and predictable, even a random walk produces reasonable forecasts, and the marginal contribution of ML sophistication shrinks.

This asymmetry has a practical implication: the value of ML for volatility forecasting is concentrated in precisely the periods when accurate forecasts matter most (high-volatility environments where risk management decisions have the largest impact on portfolio outcomes).

6 Robustness

We examine the robustness of our main findings along two dimensions: subperiod stability and statistical significance of forecast differences.

6.1 Subperiod Analysis

To test whether our rankings are driven by a specific market episode, we split the test period into two subsamples by volatility regime:

- **High volatility (2022):** The bear market year, with S&P 500 down 19.4% and 5-day realized volatility frequently exceeding 20% annualized. This subsample contains 251 trading days.
- **Lower volatility (2023–2025):** The recovery and rate-normalization period, with 5-day RV mostly below 15% annualized. This subsample contains 729 trading days.

Table 6: QLIKE by subperiod (lower is better).

Model	**2022 (high vol)**	**2023–2025 (lower vol)**
GARCH(1,1)	Higher	Lower
GJR-GARCH	Best individual	Competitive
LightGBM	Competitive	Competitive
HAR-RV	Weakest	Competitive
Ensemble	Best overall	Best overall

The key finding is that model rankings are broadly stable across regimes, but the *magnitude* of differences is larger in 2022. In the high-volatility subsample, GJR-GARCH's advantage over standard GARCH is pronounced—the leverage effect matters most when markets are falling sharply. In the lower-volatility period, all models converge, because stable volatility is easy to forecast regardless of methodology. The ensemble maintains its lead in both subsamples, confirming that forecast combination is a robust strategy across regimes.

6.2 Statistical Significance

We test whether the QLIKE differences between key model pairs are statistically significant using the Diebold and Mariano (1995) test, applied to the loss differential series

$d_t = L(\sigma_t, \hat{\sigma}_t^A) - L(\sigma_t, \hat{\sigma}_t^B)$ where L is the QLIKE loss. The null hypothesis is equal expected loss.

Over the full test period ($N = 980$), the following pairwise comparisons are informative:

- **GJR-GARCH vs. GARCH(1,1):** The asymmetric specification significantly outperforms the symmetric one ($p < 0.05$), confirming that the leverage effect is statistically meaningful, not just economically meaningful.
- **Ensemble vs. GARCH(1,1):** The ensemble significantly outperforms standard GARCH ($p < 0.05$).
- **LightGBM vs. HAR-RV:** LightGBM outperforms HAR-RV, though the difference is significant only at the 10% level, reflecting HAR-RV's competitiveness despite its simplicity.
- **Ensemble vs. GJR-GARCH:** The ensemble edges out GJR-GARCH on QLIKE, but the difference is not statistically significant ($p > 0.10$). The two models are statistically indistinguishable over this test period.

The practical implication is clear: if forced to choose a single model, GJR-GARCH is a defensible choice. If combining models is feasible (as it usually is), the ensemble is weakly preferred.

6.3 Limitations and Extensions

We note three limitations that future work should address. First, we evaluate only the 5-day forecast horizon; extending to 1-day and 22-day horizons would test whether GARCH's relative advantage persists at shorter horizons where its one-step-ahead structure is most natural. Second, our comparison is limited to the S&P 500; results may differ for individual stocks, emerging markets, or other asset classes where the leverage effect has different dynamics. Third, we exclude recurrent neural networks (LSTM, GRU) from our final comparison due to training instabilities encountered during our implementation; a more careful neural network implementation with alternative training procedures may yield different conclusions.

7 Conclusion

This paper ran a controlled horse race between eight volatility forecasting models on S&P 500 data from 2004 through 2025, with out-of-sample evaluation over the 2022–2025 period. The results can be summarized in three sentences. GJR-GARCH is the best individual model, because the leverage effect—negative returns driving volatility up more than positive returns of the same magnitude—is the single most valuable signal in a test period dominated by asymmetric drawdowns. Tree-based methods (LightGBM, XGBoost) are strong runners-up that benefit from conditioning on VIX and multi-horizon features. A simple ensemble of LightGBM, HAR-RV, and GARCH beats everything on QLIKE.

A few reflections that go beyond the numbers.

First, the strength of GJR-GARCH is a reminder that a well-specified parametric model can beat machine learning when the parametric assumption happens to match the data-generating process. The leverage effect is real, persistent, and well-captured by the GJR asymmetric term. ML models can learn this asymmetry from data (the "negative return" features in our tree models serve a similar purpose), but they have to discover it from noisy examples rather than encoding it structurally. In this particular test period—heavy on asymmetric downside risk—the parametric advantage won.

Second, the underperformance of HAR-RV relative to our expectations warrants comment. HAR has been a perennial strong performer in the volatility forecasting literature, often matching or beating ML models. Our finding that it ranks last on QLIKE is specific to the 2022–2025 test period, which is unusually rich in regime changes. HAR conditions only on past realized volatility and has no access to VIX (a forward-looking, market-implied signal) or to asymmetric response terms. In calmer markets, we suspect HAR would be more competitive.

Third, the ensemble result is the most practically useful finding. A risk manager who maintains three structurally diverse models and averages their forecasts will achieve the best QLIKE without needing to select a single "winner." This approach is robust to model misspecification and insulates against the worst case where any individual model fails in an unfamiliar regime.

We close with limitations. Our comparison is limited to a single asset (the S&P 500), and the results may not generalize to individual stocks, emerging markets, or other asset classes. We use daily data rather than intraday data; a HAR model with 5-minute realized volatility would likely outperform our daily version. We do not test transformer architectures, which have shown promise in recent work but require substantially more data and compute to train. Finally, our test period, while deliberately challenging, is still only four years long. A longer out-of-sample period would provide stronger evidence, but we are limited by when the models were trained.

All code, data, and results used in this paper are available at [GitHub repository URL]. We encourage replication, extension, and correction.

References

Andersen, T. G., Bollerslev, T., Christoffersen, P. F., and Diebold, F. X. (2006). Volatility and correlation forecasting. In *Handbook of Economic Forecasting*, volume 1, pages 777–878. Elsevier.

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.

Bollerslev, T., Patton, A. J., and Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18.

Bucci, A. (2020). Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD*, pages 785–794.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Garman, M. B. and Klass, M. J. (1980). On the estimation of security price volatilities from historical data. *The Journal of Business*, 53(1):67–78.

Glosten, L. R., Jagannathan, R., and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801.

Hansen, P. R. and Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7):873–889.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30.

Khan, A. (2026). Machine learning in quantitative finance: A systematic review of methods, applications, and open challenges (2015–2025). Technical Report 6562398, SSRN.

Kim, H. Y. and Won, C. H. (2019). Forecasting financial volatility with attention-based neural networks. *Expert Systems with Applications*, 122:40–50.

Luong, C. and Dokuchaev, N. (2021). Forecasting realized volatility with machine learning. *Journal of Risk and Financial Management*, 14(4):171.

Mittnik, S., Robinzonov, N., and Spindler, M. (2022). Machine learning for volatility forecasting. *Journal of Financial Econometrics*.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370.

Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *The Journal of Business*, 53(1):61–65.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256.

Patton, A. J. (2019). Comparing possibly misspecified forecasts. *Journal of Business & Economic Statistics*, 38(4):796–809.

Patton, A. J. and Sheppard, K. (2015). Good volatility, bad volatility: Signed jumps and the persistence of volatility. *The Review of Economics and Statistics*, 97(3):683–697.

Poon, S.-H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature*, 41(2):478–539.

Rahimikia, E. and Poon, S.-H. (2021). Machine learning for realized volatility forecasting. *Journal of Financial Econometrics*.

Sheppard, K. (2023). arch: Autoregressive conditional heteroskedasticity models in Python. https://github.com/bashtage/arch. Version 6.x.

Timmermann, A. (2006). Forecast combinations. *Handbook of Economic Forecasting*, 1:135–196.